10 Fig. 10.1 shows the path of the Earth as it orbits the Sun. X is a position on the Earth where scientists observe the apparent motion of the Sun throughout the year.

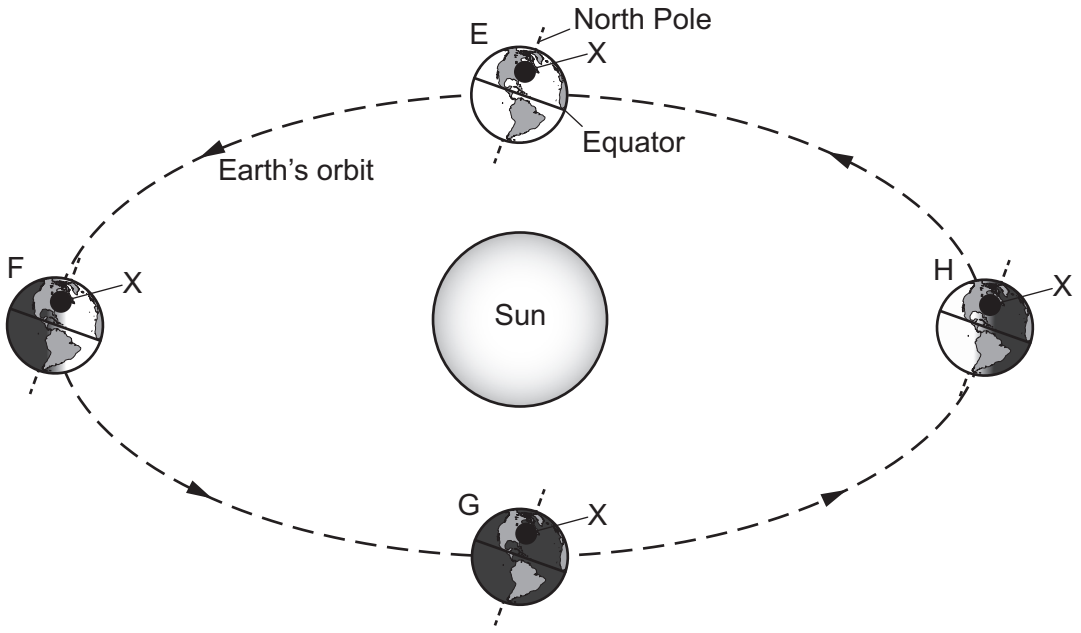


Fig. 10.1

(a) Determine how many days it takes the Earth to move around its orbit from F to G. Explain your answer.

number of days = ...

explanation .....

.....

[2]

(b) Fig. 10.1 shows four positions E, F, G and H of the Earth in its orbit of the Sun.

(i) Identify the position of the Earth when it is summer at X. ..... [1]

(ii) Identify the position of the Earth when it is winter at X. ..... [1]

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(c) The orbital speed of the Earth around the Sun is approximately 3.0×10^4 m/s.

Calculate the average radius of the Earth's orbit.

radius = ... [3]

(d) Earth is a planet in the Solar System. State **one other** type of naturally occurring object that is present in the Solar System.

..... [1]

[Total: 8]

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